Digital design intern ADI

ASIC flow for 4bit Counter

Made by:

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The RTL code:

```
// counter4.v
module counter4 (
    input  wire       clk,
    input  wire       rst_n,
    input  wire       en,
    output reg  [3:0] count
);

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n)
            count <= 4'b0000;
        else if (en)
            count <= count + 1'b1;
    end

endmodule
```

Configure file

```
set ::env(DESIGN_NAME) "counter4"

set ::env(VERILOG_FILES) "$::env(DESIGN_DIR)/src/counter4.v"

set ::env(CLOCK_PORT) "clk"
set ::env(CLOCK_NET)  "clk"
set ::env(CLOCK_PERIOD) "10"

set ::env(FP_SIZING) "absolute"
set ::env(DIE_AREA) "0 0 100 100"

set ::env(PL_TARGET_DENSITY) "0.5"

set ::env(SYNTH_STRATEGY) "AREA 0"
```

Results

Timing:

```
worst_slack

===========================================================================
 report_worst_slack -max (Setup)
============================================================================
worst slack 6.77

===========================================================================
 report_worst_slack -min (Hold)
============================================================================
worst slack 0.23
worst_slack_end
clock_skew

===========================================================================
 report_clock_skew
============================================================================
Clock clk
Latency      CRPR       Skew
_21_/CLK ^
   0.29
_22_/CLK ^
   0.27     -0.02       0.01

clock_skew_end
power_report

===========================================================================
 report_power
============================================================================
Group                  Internal  Switching    Leakage      Total
                          Power      Power      Power      Power (Watts)
----------------------------------------------------------------
Sequential             1.72e-05   1.49e-06   3.50e-11   1.87e-05  21.5%
Combinational          5.35e-05   1.49e-05   2.66e-10   6.84e-05  78.5%
Macro                  0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
Pad                    0.00e+00   0.00e+00   0.00e+00   0.00e+00   0.0%
----------------------------------------------------------------
Total                  7.07e-05   1.64e-05   3.01e-10   8.71e-05 100.0%
                          81.1%      18.9%       0.0%
power_report_end
area_report

===========================================================================
 report_design_area
============================================================================
Design area 593 u^2 9% utilization.
area_report_end
```

DRCs:

```
    ubm
Scaled tech values by 2 / 1 to match internal grid scaling
Loading sky130A Device Generator Menu ...
Using technology "sky130A", version 1.0.291-36-g41c0908
Warning: Calma reading is not undoable!  I hope that's OK.
Library written using GDS-II Release 3.0
Library name: counter4
Reading "sky130_fd_sc_hd__decap_8".
Reading "sky130_ef_sc_hd__decap_12".
Reading "sky130_fd_sc_hd__decap_3".
Reading "sky130_fd_sc_hd__buf_2".
Reading "sky130_fd_sc_hd__fill_1".
Reading "sky130_fd_sc_hd__tapvpwrvgnd_1".
Reading "sky130_fd_sc_hd__decap_4".
Reading "sky130_fd_sc_hd__decap_6".
Reading "sky130_fd_sc_hd__clkbuf_2".
Reading "sky130_fd_sc_hd__and2_1".
Reading "sky130_fd_sc_hd__xnor2_1".
Reading "sky130_fd_sc_hd__fill_2".
Reading "sky130_fd_sc_hd__clkbuf_16".
Reading "sky130_fd_sc_hd__a31o_1".
Reading "sky130_fd_sc_hd__or2_1".
Reading "sky130_fd_sc_hd__clkbuf_1".
Reading "sky130_fd_sc_hd__dfrtp_1".
Reading "sky130_fd_sc_hd__and4_1".
Reading "sky130_fd_sc_hd__nand2_1".
Reading "sky130_fd_sc_hd__and2b_1".
Reading "sky130_fd_sc_hd__xor2_1".
Reading "counter4".
[INFO]: Loading counter4

DRC style is now "drc(full)"
Loading DRC CIF style.
No errors found.
[INFO]: COUNT: 0
[INFO]: Should be divided by 3 or 4
[INFO]: DRC Checking DONE (/home/opentools/OpenLane/designs/counter4/runs/RUN_2026.08.19_23.06.21/reports/signoff/drc.rpt)
[INFO]: Saving mag view with DRC errors (/home/opentools/OpenLane/designs/counter4/runs/RUN_2026.08.19_23.06.21/results/signoff/counter4.drc.mag)
[INFO]: Saved
```

Plain Text Tab Width: 8 Ln 1, Col 1 INS

se pointer inside or press Ctrl+G.

LVS:

warnings.log errors.log cmds.log drc.tcl drc.klayout.xml drc.rdb drc.rpt drc.tr detailed.drc 27-drc.log 26-lef.log 26-counter4.lvs.lef.log

```
LVS reports no net, device, pin, or property mismatches.

Total errors = 0
```

```
  Class: sky130_fd_sc_hd__fill_1 instances:   1
  Class: sky130_fd_sc_hd__fill_2 instances:   1
Circuit contains 28 nets.

Circuit 1 contains 31 devices, Circuit 2 contains 31 devices.
Circuit 1 contains 28 nets,    Circuit 2 contains 28 nets.

Netlists match uniquely.
Result: Circuits match uniquely.
Logging to file "/home/opentools/OpenLane/designs/counter4/runs/RUN_2026.08.19_23.06.21/logs/signoff/26-counter4.lef.log" disabled
LVS Done.
```

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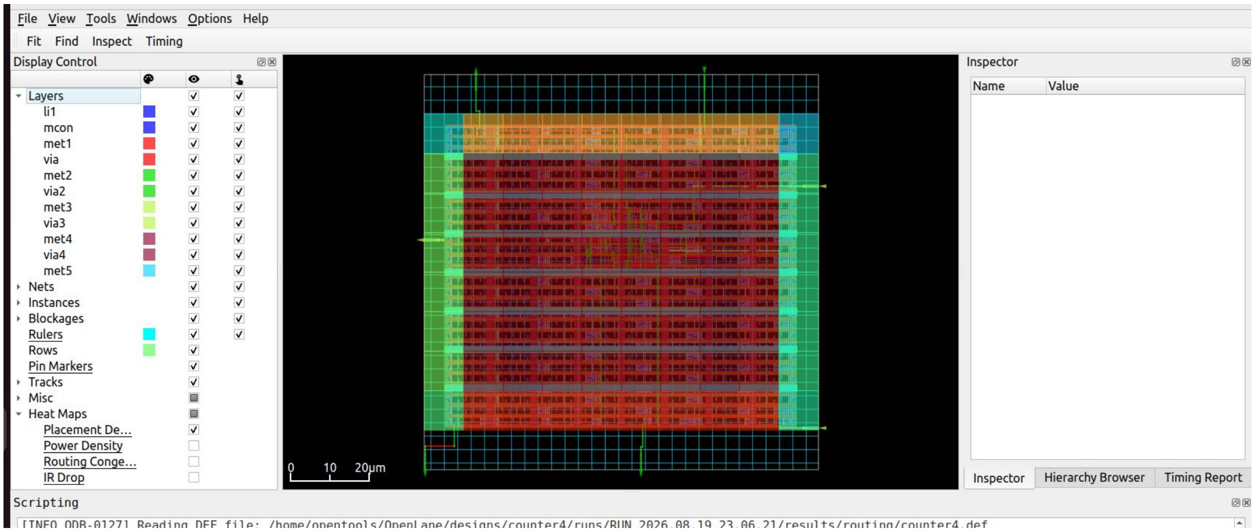
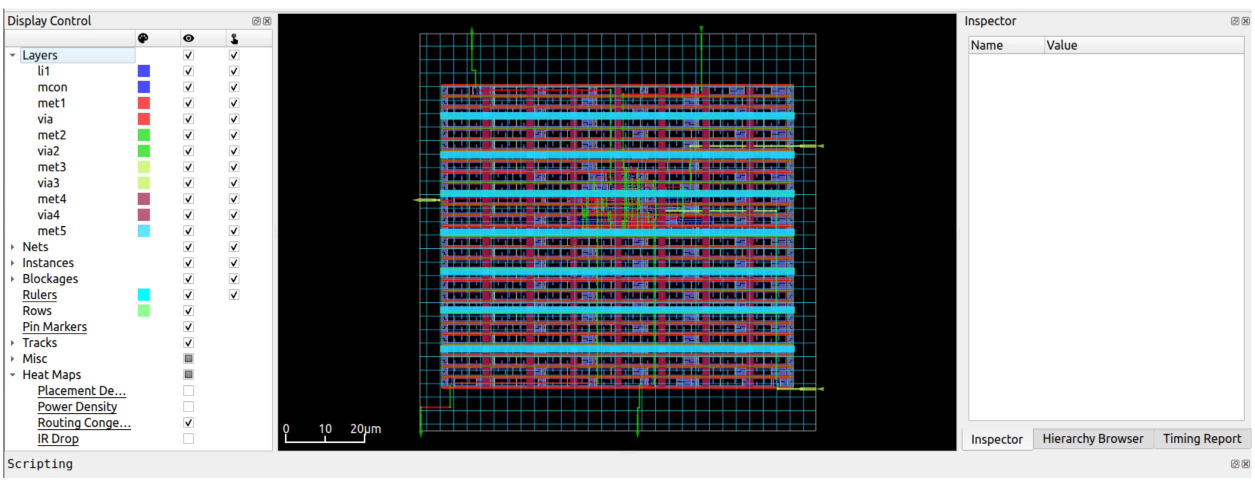
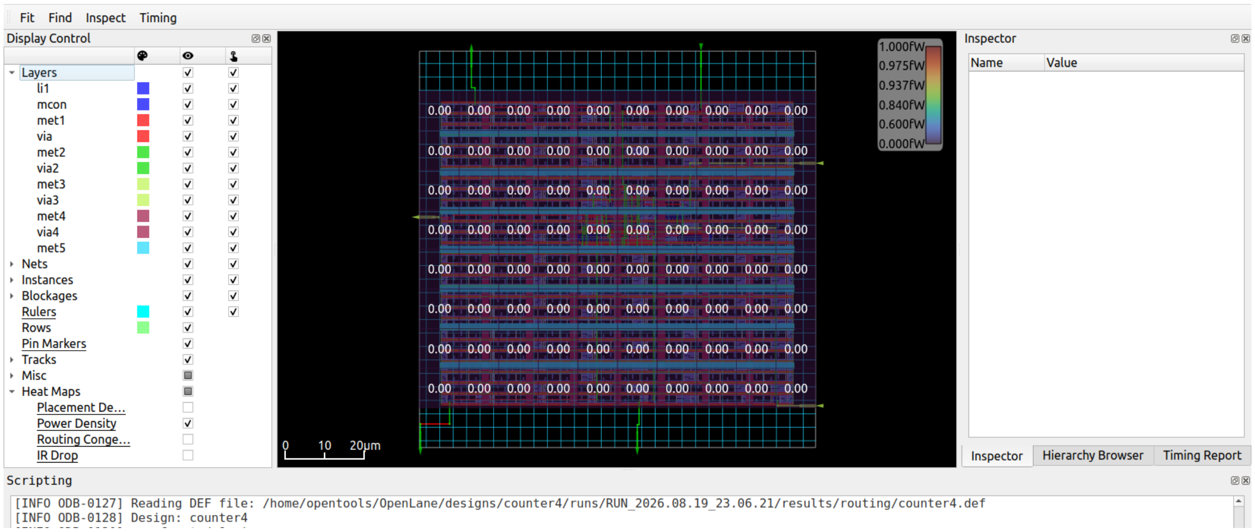
```
[INFO]: Writing Verilog...
[STEP 11]
[INFO]: Running Detailed Placement...
% run_cts
[STEP 12]
[INFO]: Running Clock Tree Synthesis...
[STEP 13]
[INFO]: Writing Verilog...
% run_routing
[INFO]: Routing...
[STEP 14]
[INFO]: Running Global Routing Resizer Timing Optimizations...
[STEP 15]
[INFO]: Writing Verilog...
[STEP 16]
[INFO]: Running Detailed Placement...
[STEP 17]
[INFO]: Running Global Routing...
[INFO]: Starting FastRoute Antenna Repair Iterations...
[STEP 18]
[INFO]: Running Fill Insertion...
[STEP 19]
[INFO]: Writing Verilog...
[STEP 20]
[INFO]: Running Detailed Routing...
[INFO]: No DRC violations after detailed routing.
[STEP 21]
[INFO]: Writing Verilog...
1787170387
% run_magic
[STEP 22]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDS-II with Magic...
[INFO]: Generating MAGLEF views...
% run_magic_spice_export
[STEP 23]
[INFO]: Running Magic Spice Export from LEF...
% run_lvs
[STEP 24]
[INFO]: Writing Powered Verilog...
[STEP 25]
[INFO]: Writing Verilog...
[STEP 26]
[INFO]: Running LEF LVS...
% run_magic_drc
[STEP 27]
[INFO]: Running Magic DRC...
[INFO]: Converting Magic DRC Violations to Magic Readable Format...
[INFO]: Converting Magic DRC Violations to Klayout XML Database...
[INFO]: No DRC violations after GDS streaming out.
%
```

Placement congestion



Routing congestion




Fit Find Inspect Timing
Display Control
Layers
li1
mcon
met1
via
met2
via2
met3
via3
met4
via4
met5
Nets
Instances
Blockages
Rulers
Rows
Pin Markers
Tracks
Misc
Heat Maps
Placement De...
Power Density
Routing Conge...
IR Drop
1.000fW
0.975fW
0.937fW
0.840fW
0.600fW
0.000fW
0 10 20μm
Inspector
Name
Value
Inspector
Hierarchy Browser
Timing Report
Scripting
[INFO ODB-0127] Reading DEF file: /home/opentools/OpenLane/designs/counter4/runs/RUN_2026.08.19_23.06.21/results/routing/counter4.def
[INFO ODB-0128] Design: counter4

Final GDS: